

Session 2: Bayesian regression with Big Data

Modern Macroeconometrics

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OVERVIEW OF SESSION 2

1. **Kick off this session with some (light) Bayesian basics:** Provide a brief overview of Bayes' Theorem, Bayesian learning and updating, prior and posterior distributions, and predictive inference
2. **Bayesian regression model:** See how general Bayesian theory works in a familiar regression model (connection to classic estimators like OLS)
3. **ML-flavoured shrinkage priors for Big Data:** Stochastic search variable selection (SSVS), Bayesian LASSO (and its frequentist duality), and global-local shrinkage priors
4. **Application:** Which variables predict US GDP growth? A first look at growth at risk, in the spirit of Plagborg-Møller et al. (2020, *Brookings Papers*)

GENERAL HEALTH WARNING FOR THE WHOLE COURSE

To keep us on time (and save you from too much technical detail), I skip over most of the Bayesian computational nitty-gritty

Bayesian basics

WHAT DOES A BAYESIAN (MACRO-)ECONOMETRICIAN DO?

- An econometrician mainly does three things
 1. Estimation of unknown parameters θ (e.g., regression coefficients) from data y
 2. Prediction
 3. Variable selection (e.g., hypothesis testing)
- Bayesian econometrics does these based on a few simple rules of probability in a single coherent framework
 - Begin with general concepts in Bayesian theory before getting to specific models
 - If you know these general concepts you will never get lost
- Bayesian methods for freeing up classical assumptions are relatively straightforward (some of them we will discuss in the next sessions)

CONDITIONAL PROBABILITIES AND BAYES' THEOREM

- The **conditional probability** of A given B , denoted $\Pr(A \mid B)$, is the probability of event A occurring given that event B has occurred
- For two events A and B , the rules of conditional probability are:

$$\Pr(A \mid B) = \frac{\Pr(A, B)}{\Pr(B)} \quad \Pr(B \mid A) = \frac{\Pr(A, B)}{\Pr(A)}$$

- Combining these two yields **Bayes' theorem**:

$$\Pr(B \mid A) = \frac{\Pr(A \mid B) \Pr(B)}{\Pr(A)}$$

- This may sound abstract, so let's illustrate this using a data/parameter context

LEARNING ABOUT PARAMETERS IN A GIVEN MODEL (ESTIMATION)

- Assume a single model which depends on parameter θ
- Want to figure out properties of the posterior $p(\theta | \mathbf{y})$
- It is convenient to use Bayes' rule to write the posterior in a different way
- Bayes' rule lies at the heart of Bayesian econometrics:

$$p(B | A) = \frac{p(A | B)p(B)}{p(A)}$$

- Replace B by θ and A by $\mathbf{y}$ to obtain:

$$p(\theta | \mathbf{y}) = \frac{p(\mathbf{y} | \theta)p(\theta)}{p(\mathbf{y})}$$

BAYES' THEOREM: POSTERIOR $\propto$ LIKELIHOOD $\times$ PRIOR

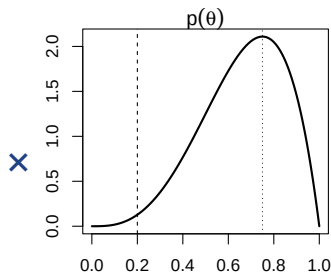
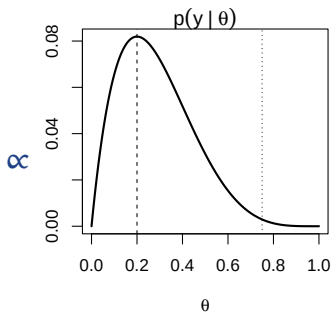
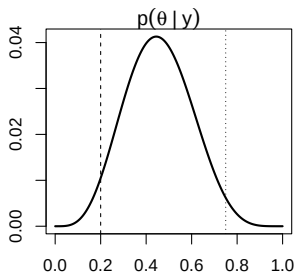
- For estimation can ignore the term $p(\mathbf{y})$ since it does not involve θ
- For data $\mathbf{y}$ and unknown parameter θ , Bayes' rule gives:

$$\underbrace{p(\theta | \mathbf{y})}_{\text{Posterior}} \propto \underbrace{p(\mathbf{y} | \theta)}_{\text{Likelihood}} \times \underbrace{p(\theta)}_{\text{Prior}}$$

- Bayesians treat $p(\theta | \mathbf{y})$ as being of fundamental interest: “Given the data, what do we know about θ ?”
- Treatment of θ as a random variable is controversial among some econometricians
- **Frequentist econometrics** says that θ is not a random variable

BAYES' THEOREM IN ACTION

We combine data (likelihood) and non-data (prior) information to obtain an updated (proportional) posterior distribution



PREDICTION IN A SINGLE MODEL

- Prediction based on the **predictive density** $p(y^* | \mathbf{y})$
- Since a marginal density can be obtained from a joint density through integration:

$$p(y^* | \mathbf{y}) = \int p(y^*, \theta | \mathbf{y}) d\theta$$

- Term inside integral can be rewritten as:

$$p(y^* | \mathbf{y}) = \int p(y^* | \mathbf{y}, \theta) p(\theta | \mathbf{y}) d\theta$$

- Prediction involves the posterior; integrates out parameter uncertainty of θ

UNDERSTANDING THE PRIOR

- The prior $p(\theta)$ is the controversial aspect of Bayesian inference; it sounds unscientific to introduce non-data information
- Bayesian answers:
 1. Often we do have prior information and, if so, we should include it (more information is good)
 2. Can work with so-called “noninformative” priors or a prior that recovers frequentist estimators/properties (see, e.g., LASSO)
 3. Can use hierarchical priors which treat prior hyperparameters as parameters and estimate them; the prior “learns” from the data (see shrinkage priors later on)
 4. Priors obtained from a theoretical model (e.g., a DSGE model)
 5. Prior sensitivity analysis to assess robustness

BAYESIAN COMPUTATION

- Before jumping into the Bayesian regression model, how do we present results from a Bayesian empirical analysis?
- $p(\theta \mid \mathbf{y})$ is a pdf; especially if θ is a vector of many parameters cannot present a graph of it
- Want features analogous to frequentist point estimates and confidence intervals
- All these features have integrals in them

IMPORTANT POSTERIOR FEATURES

- › All of these posterior features have the form

$$\mathbb{E}[g(\theta) | \mathbf{y}] = \int g(\theta)p(\theta | \mathbf{y})d\theta,$$

where $g(\theta)$ is a **function of interest**

- › A common point estimate is the mean of the posterior density (or **posterior mean**), obtained with $g(\theta) = \theta$
- › Common measure of dispersion is the **posterior standard deviation** (square root of **posterior variance**), with $\mathbb{V}(\theta | \mathbf{y}) = \mathbb{E}[\theta^2 | \mathbf{y}] - \mathbb{E}[\theta | \mathbf{y}]^2$
- › The integrals involved in Bayesian analysis are usually evaluated using **simulation methods**

INTUITION OF POSTERIOR SIMULATION

- Frequentist asymptotic theory uses Laws of Large Numbers (LLN) and Central Limit Theorems (CLT)
- A typical LLN: “consider a random sample, $Y_1, \dots, Y_N$, as N goes to infinity, the average converges to its expectation” (e.g. $\bar{Y} \rightarrow \mu$)
- Bayesians use LLN: “consider a random sample from the posterior, $\theta^{(1)}, \dots, \theta^{(S)}$, as S goes to infinity, the average of these converges to $\mathbb{E}[\theta \mid y]$ ”

EXAMPLE: MONTE CARLO INTEGRATION

- › Let $\theta^{(s)}$ for $s = 1, \dots, S$ be a random sample from $p(\theta \mid \mathbf{y})$ and define

$$\widehat{g}_S = \frac{1}{S} \sum_{s=1}^S g\left(\theta^{(s)}\right),$$

then $\widehat{g}_S$ converges to $\mathbb{E}[g(\theta) \mid \mathbf{y}]$ as S goes to infinity

› Computational trade-offs

- › Monte Carlo integration approximates $\mathbb{E}[g(\theta) \mid \mathbf{y}]$, but only if S were infinite would the approximation error be zero
- › We can choose any value for S but larger values of S will increase computational burden
- › To gauge size of approximation error, use a CLT to obtain numerical standard error

Bayesian linear regression model

HOW TO APPROACH A REGRESSION PROBLEM AS A BAYESIAN

- › Suppose we want to estimate a simple AR(1) model:

$$y_t = \rho y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2)$$

- › We have data $\mathbf{y} = (y_1, \dots, y_T)'$ and two unknown model parameters (ρ and σ^2)
- › Assume we know that y_t is a stationary time series (e.g., a growth rate)
- › What else do we know about the data? (e.g., how is the data distributed?)
- › Given this information, what prior beliefs can we formulate about the AR(1) parameter ρ and the variance parameter σ^2 ?
- › In other words, we wish to specify the prior distributions $p(\rho)$ and $p(\sigma^2)$
 - › What do we know about ρ for certain?
 - › What do we know about σ^2 for certain?

A REGRESSION MODEL

- ▶ Let's assume K explanatory variables $x_{t1}, \dots, x_{tK}$ for $t = 1, \dots, T$, with $x_{t1} = 1$ for an intercept:

$$y_t = \beta_1 + \beta_2 x_{t2} + \dots + \beta_K x_{tK} + \varepsilon_t$$

- ▶ Stacking over $t = 1, \dots, T$ gives $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, with

$$\underset{T \times 1}{\mathbf{y}} = \begin{bmatrix} y_1 \\ \vdots \\ y_T \end{bmatrix}, \quad \underset{T \times K}{\mathbf{X}} = \begin{bmatrix} 1 & x_{12} & \cdots & x_{1K} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{T2} & \cdots & x_{TK} \end{bmatrix}, \quad \underset{K \times 1}{\boldsymbol{\beta}} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_K \end{bmatrix}, \quad \underset{T \times 1}{\boldsymbol{\varepsilon}} = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_T \end{bmatrix}$$

- ▶ Classical assumption: $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}_T, \sigma^2 \mathbf{I}_T)$

FREQUENTIST VS BAYESIAN: TWO VIEWS OF THE SAME PROBLEM

- **Frequentist:** β is a fixed unknown constant (the **truth**); y , and hence $\hat{\beta}_{OLS}$, is random, so inference relies on **CLT**:

$$\underbrace{\hat{\beta}_{OLS}}_{\text{random}} = (X'X)^{-1}X' \underbrace{y}_{\text{random}} = \underbrace{\beta}_{\text{fixed (true)}} + (X'X)^{-1}X'\varepsilon$$
$$\xrightarrow{\text{CLT}} \hat{\beta}_{OLS} \sim \mathcal{N}\left(\beta, \sigma^2(X'X)^{-1}\right)$$

- **Bayesian:** β is a random variable; combine prior $p(\beta)$ with likelihood $p(y | \beta)$ to get posterior $p(\beta | y)$; inference via the posterior distribution
- **Key connection:** With a noninformative prior $p(\beta) \propto \text{const}$, the Bayesian posterior mean = $\hat{\beta}_{OLS}$

PARAMETERS OF INTEREST AND THE BIG DATA CHALLENGE

- › Parameters of interest: $K \times 1$ coefficient vector β and the error variance σ^2
- › OLS: $\hat{\beta} = (X'X)^{-1}X'y$
- › In the **Big Data** context K can be large; comparable to or even larger than T
- › OLS **fails** when $K \geq T$: $X'X$ is not invertible
- › **Bayesian view**: Incorporate prior and shrinkage information when K is large

THE LIKELIHOOD FUNCTION

- › Likelihood can be derived under the classical assumptions
- › All elements of X are fixed (i.e., not random variables)
- › OLS quantities given by:

$$\hat{\beta} = (X'X)^{-1} X'y, \quad \nu = T - K, \quad s^2 = \frac{(y - X\hat{\beta})' (y - X\hat{\beta})}{\nu}$$

- › Likelihood function can be written in terms of OLS quantities

$$p(y \mid \beta, \sigma^2) = \frac{1}{(2\pi)^{\frac{T}{2}}} \underbrace{\exp \left[-\frac{1}{2\sigma^2} (\beta - \hat{\beta})' X'X (\beta - \hat{\beta}) \right]}_{\text{multivariate Normal in } \beta} \times \underbrace{(\sigma^2)^{-\frac{T}{2}} \exp \left[-\frac{1}{2\sigma^2} \nu s^2 \right]}_{\text{inverse Gamma in } \sigma^2}$$

Choosing a prior

NATURAL CONJUGATE PRIOR

- › Common starting point is natural conjugate prior
- › β conditional on σ^2 is **multivariate Normal** and prior for error variance σ^2 is **inverse Gamma**:

$$\beta \mid \sigma^2 \sim \mathcal{N}(\underline{\beta}, \sigma^2 \underline{V})$$

$$\sigma^2 \sim \mathcal{IG}(\underline{a}, \underline{b})$$

- › **Hyperparameters** chosen by the researcher:
 - › $\underline{\beta}$: Prior mean for the coefficients
 - › $\underline{V}$: Prior covariance matrix (scales with σ^2)
 - › $\underline{a}$: Prior shape parameter for the error variance
 - › $\underline{b}$: Prior scale parameter for the error variance

THE JOINT POSTERIOR OF β AND σ^2

- › Multiplying likelihood by prior and collecting terms, the posterior can be shown to have the same form as the prior
- › The conditional posterior for β (conditional on σ^2) is **multivariate Normal** and the posterior for error variance σ^2 is **inverse Gamma**:

$$\beta \mid y, \sigma^2 \sim \mathcal{N}(\bar{\beta}, \sigma^2 \bar{V})$$

with

$$\begin{aligned}\bar{V} &= (\underline{V}^{-1} + X'X)^{-1} \\ \bar{\beta} &= \bar{V} (\underline{V}^{-1} \underline{\beta} + X'X \hat{\beta})\end{aligned}$$

$$\sigma^2 \mid y \sim \mathcal{IG}(\bar{a}, \bar{b})$$

with

$$\begin{aligned}\bar{a} &= \underline{a} + \frac{T}{2} \\ \bar{b} &= \underline{b} + \frac{1}{2}(y'y + \underline{\beta}'\underline{V}^{-1}\underline{\beta} - \bar{\beta}'\bar{V}^{-1}\bar{\beta})\end{aligned}$$

THE MARGINAL POSTERIOR OF β AND PREDICTIVE DENSITY

- The marginal posterior for β is obtained by integrating out σ^2
- It turns out that β is multivariate t -distributed:

$$\beta \mid y \sim t \left(\bar{\beta}, \frac{\bar{b} \bar{V}}{\bar{a}}, 2\bar{a} \right)$$

- **Predictive density** for new observation y^* given regressors x^* :

$$p(y^* \mid x^*, y) = \int p(y^* \mid x^*, \beta, \sigma^2) p(\beta, \sigma^2 \mid y) d\beta d\sigma^2$$

- It turns out the predictive density is also a t distribution, automatically capturing parameter uncertainty in the forecast

WHAT DOES A PRIOR DO?

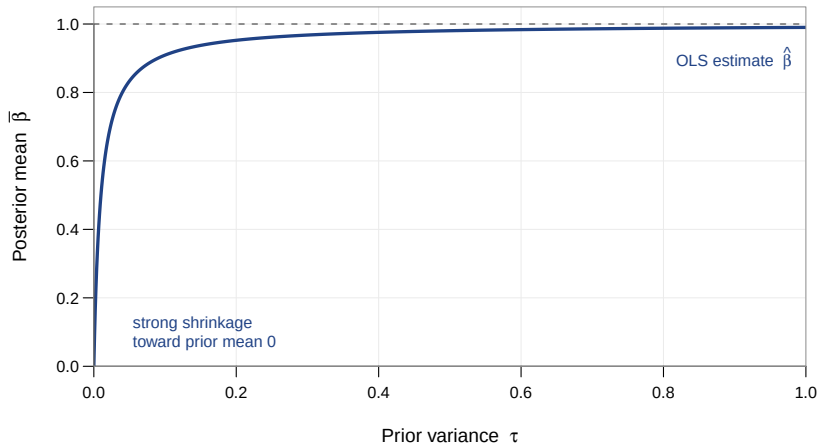
- To understand the idea of shrinkage, let's illustrate what the prior does
 - $\sigma^2 = 1$ is known and β is a scalar
 - $X = (1, \dots, 1)'$, such that $X'X = T$
 - $\underline{V} = \tau$ and $\underline{\beta} = 0$
- In this case the posterior is:

$$\bar{\beta} = \frac{T}{\frac{1}{\tau} + T} \hat{\beta}$$

- Think of the extreme cases ($\tau \rightarrow \infty$ and $\tau \rightarrow 0$)
 - If $\tau \rightarrow \infty$, then $\frac{1}{\tau} \rightarrow 0$ and $\frac{T}{\frac{1}{\tau} + T} \rightarrow 1$, so $\bar{\beta} \rightarrow \hat{\beta}$ (i.e., **OLS**)
 - If $\tau \rightarrow 0$, then $\frac{1}{\tau} \rightarrow \infty$ and $\frac{T}{\frac{1}{\tau} + T} \rightarrow 0$, so $\bar{\beta} \rightarrow 0$ (the prior mean, i.e., **exclusion**)

IMPLIED PRIOR SHRINKAGE BY VARYING τ

$$\bar{\beta} = \frac{T}{1/\tau + T} \hat{\beta}, \text{ for } T = 100 \text{ and } \hat{\beta} = 1$$



PRIOR SHRINKAGE

- Posterior mean is pulled towards zero (“shrinkage”)
- Commonly done to avoid over-fitting/over-parameterisation problems
- Strength of prior shrinkage controlled through prior variance
 - If τ is small, then strong prior information β is near 0
 - If τ is big, then prior becomes more noninformative [see Appendix](#)
- Note: exactly what “small” and “large” means depends on the empirical application and units of measurement of data

Shrinkage priors for Big Data

SOME CONTEXT: BIG DATA REGRESSION PROBLEM

- Re-consider a regression with K potential predictors: $y_t = x_t' \beta + \varepsilon_t$
- Most predictors are likely irrelevant, but a priori we do not know which ones (and the degree of relevance varies across them; see Session 1)
- A noninformative prior cannot discriminate between relevant and irrelevant variables
- We need priors that:
 - **Shrink some irrelevant predictors towards zero** to avoid overfitting
 - **Keep some relevant predictors close to OLS** to preserve signal

SOME MOTIVATION: GROWTH AT RISK FOR LATER ON

- **Growth at risk:** Forecast the **conditional distribution** of future GDP growth, especially its **lower tail** (downside risk; see Session 5), from many macro-financial predictors (e.g., FRED-QD)
- **Why this is hard:** Many predictors K , short macro samples T ($K > T$); the **tail** is identified by only a few left-tail observations
- But before we turn to tail behaviour and quantiles, we first work through the standard linear regression (the conditional mean)
- Plagborg-Møller et al. (2020, *Brookings Papers*) reassess the GaR story and find that financial variables have very limited predictive power; signals are driven by a handful of episodes

INTERPRETATION OF BAYESIAN SHRINKAGE

- › The posterior mean $\bar{\beta}$ is a **weighted average** of the prior mean $\underline{\beta}$ and the OLS estimator $\hat{\beta}$:

$$\bar{\beta} = \underbrace{\bar{V}\bar{V}^{-1}}_{\text{prior weight}} \underline{\beta} + \underbrace{\bar{V}X'X}_{\text{data weight}} \hat{\beta}$$

- › Setting $\underline{\beta} = 0$ implies the posterior mean is **shrunk towards zero**
 - › Small prior variance $\bar{V}$: strong shrinkage, $\bar{\beta} \approx 0$
 - › Large prior variance $\bar{V}$: weak shrinkage, $\bar{\beta} \approx \hat{\beta}$

NON-CONJUGATE PRIORS

- › Normal linear regression with **natural conjugate prior** $\Rightarrow$ posterior and predictive distributions available in **closed form**
- › Shrinkage priors / more complex models $\Rightarrow$ **no analytical results**
- › Extensions to shrinkage priors of the regression model requiring a **Gibbs sampler** for posterior computation see Appendix
 - › **Gibbs sampler explained in a sentence**: Draw each parameter block (β , then σ^2) from its **full conditional posterior** in turn; cycling through these draws approximates the joint posterior
 - › Keep the Normal linear regression model (under the classical assumptions) as before
 - › Likelihood function same as before
 - › Parameters of model will still be β and σ^2
 - › We then only introduce flexibility through the prior

THE INDEPENDENT NORMAL-INVERSE-GAMMA PRIOR

- Before we had conjugate prior where $p(\beta \mid \sigma^2)$ was Normal density and $p(\sigma^2)$ inverse Gamma density
- Now use similar prior, but assume prior **independence** between β and σ^2

$$p(\beta, \sigma^2) = p(\beta) p(\sigma^2)$$

- $p(\beta)$ is Normal and $p(\sigma^2)$ inverse Gamma:

$$\begin{aligned}\beta &\sim \mathcal{N}(\underline{\beta}, \underline{V}) \\ \sigma^2 &\sim \mathcal{IG}(\underline{a}, \underline{b})\end{aligned}$$

- Key difference: $\underline{V}$ is now the prior covariance matrix of β directly; with the conjugate prior we had $\mathbb{V}(\beta \mid \sigma^2) = \sigma^2 \underline{V}$

THE POSTERIOR

- › Posterior $\propto$ prior $\times$ likelihood; **joint posterior** for (β, σ^2) has **no closed form**
- › But the two **conditional posteriors** are simple:

$$\beta \mid y, \sigma^2 \sim \mathcal{N}(\bar{\beta}, \bar{V})$$

with

$$\begin{aligned}\bar{V} &= \left(\underline{V}^{-1} + \sigma^{-2} \mathbf{X}'\mathbf{X} \right)^{-1} \\ \bar{\beta} &= \bar{V} \left(\underline{V}^{-1} \underline{\beta} + \sigma^{-2} \mathbf{X}'\mathbf{y} \right)\end{aligned}$$

$$\sigma^2 \mid y, \beta \sim \text{IG}(\bar{a}, \bar{b})$$

with

$$\begin{aligned}\bar{a} &= \underline{a} + \frac{T}{2} \\ \bar{b} &= \underline{b} + \frac{1}{2} (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)\end{aligned}$$

- › **Key point:** the conditionals do not directly characterize the joint posterior $\Rightarrow$ use a **Gibbs sampler** see Appendix

Popular shrinkage priors

STOCHASTIC SEARCH VARIABLE SELECTION (SSVS)

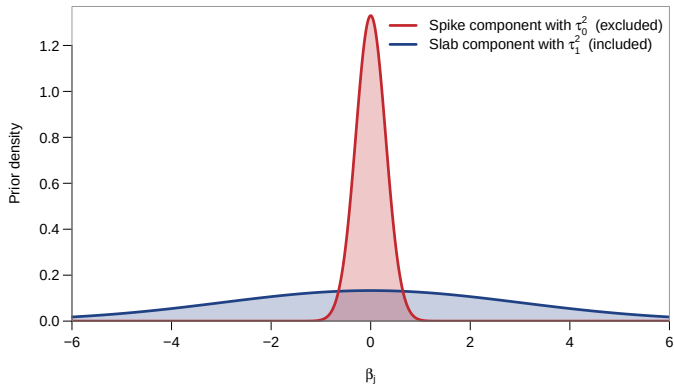
GEORGE & McCULLOCH (1993, JASA)

- A **spike and slab prior** on each coefficient β_j :

$$\beta_j \mid \gamma_j \sim (1 - \gamma_j) \mathcal{N}(0, \tau_0^2) + \gamma_j \mathcal{N}(0, \tau_1^2)$$

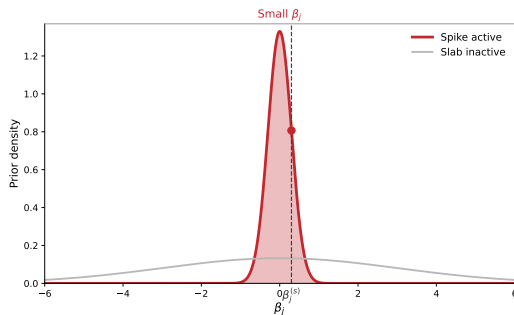
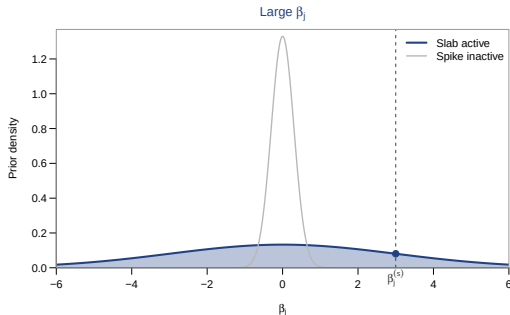
- Binary indicator $\gamma_j \in \{0, 1\}$ controls whether x_j is **included or excluded**:
 - $\gamma_j = 0$: **Spike** active (τ_0^2 very small) $\Rightarrow \beta_j \approx 0$ (variable x_j excluded)
 - $\gamma_j = 1$: **Slab** active (τ_1^2 large) $\Rightarrow \beta_j$ unconstrained (variable x_j included)
- Prior on inclusion: $\gamma_j \sim \text{Bernoulli}(\pi)$ with π typically set to 1/2 (equal prior probability of inclusion or exclusion)
- The Gibbs sampler draws β , σ^2 , and γ iteratively from well-known conditional posteriors

SSVS: THE SPIKE AND SLAB COMPONENTS ILLUSTRATED



- › **Spike** ($\gamma_j = 0$): Mass concentrated at 0 (excluded)
- › **Slab** ($\gamma_j = 1$): Diffuse/flat (included)

SSVS: WHEN IS EACH COMPONENT ACTIVE?



- At draw s : A **large** $\beta_j^{(s)}$ is explained only by the **slab** ($\gamma_j = 1$, included); a **small** $\beta_j^{(s)}$ by the **spike** ($\gamma_j = 0$, excluded)
- The Gibbs sampler updates γ_j via the inclusion probability $p(\gamma_j^{(s)} = 1 \mid \beta_j^{(s)})$

SSVS: INTERPRETATION OF POSTERIOR INCLUSION PROBABILITY

- Posterior $p(\gamma_j = 1 \mid \mathbf{y})$: **posterior inclusion probability (PIP)** tells us whether x_j matters
- Provides a natural **variable selection summary** and effectively allows for hypothesis testing
 - $\text{PIP} > 0.5$: variable j is likely relevant (**median probability model**)
 - $\text{PIP} < 0.05$: strong evidence that variable j can be excluded
- Amounts to **Bayesian model averaging (BMA)** with the posterior integrating over all 2^K possible models (accounts for model uncertainty)
- A **gold standard** for variable selection, but it breaks down in large settings ($K > 40$)

LASSO: BAYESIAN AND FREQUENTIST VIEW

- › The classic **LASSO** (Tibshirani, 1996, *JRSS:B*) minimises

$$\hat{\beta}_{\text{LASSO}} = \arg \min_{\beta} \left\{ \sum_{t=1}^T (y_t - \mathbf{x}'_t \beta)^2 + \lambda \sum_{j=1}^K |\beta_j| \right\}$$

- › **Bayesian LASSO** (Park & Casella, 2008, *JASA*): Penalty can be represented as **Laplace prior** $p(\beta_j) \propto \exp(-\lambda |\beta_j| / \sigma)$
- › The classic LASSO estimate is the **posterior mode** under this prior
- › **Bayesian** integrates over the full posterior, not just point estimate

LASSO: A HIERARCHICAL PRIOR VIEW

- › **Scale mixture representation** allows Gibbs sampling

$$\beta_j \mid \tau_j^2 \sim \mathcal{N}(0, \sigma^2 \tau_j^2), \quad \tau_j^2 \sim \text{Exp}(\lambda^2/2)$$

with each τ_j^2 being a **local** (predictor-specific) prior variance

- › As we should know by now, a small τ_j^2 shrinks β_j towards zero
- › **But:** LASSO tends to over-shrink signals
- › Motivates more sophisticated, so-called **global-local shrinkage priors**
- › Key idea is nicely summarised in the title of the paper “*Shrink Globally, Act Locally*,” Polson and Scott (2011)

GLOBAL-LOCAL SHRINKAGE PRIORS

- Modern shrinkage priors use **global** and **local** scales:

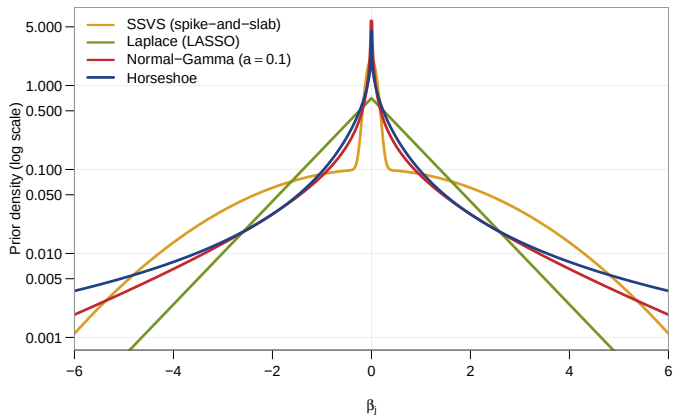
$$\beta_j \mid \phi_j^2, \lambda^2 \sim \mathcal{N}(0, \phi_j^2 \lambda^2)$$

- $\lambda > 0$: **Global shrinkage parameter** controls overall level of shrinkage
- $\phi_j > 0$: **Local shrinkage parameter** allows individual coefficients to escape the global shrinkage
- A good prior should **heavily shrink most coefficients** but **not overshrink the few large ones**
- **Shrinkage factor** $\kappa_j = \frac{1}{1+\phi_j^2 \lambda^2} \in (0, 1)$
 - **Noise** ($|\beta_j| \approx 0$): $\kappa_j \rightarrow 1$, full shrinkage to zero
 - **Signal** ($|\beta_j|$ large): $\kappa_j \rightarrow 0$, (nearly) unshrunk

GLOBAL-LOCAL SHRINKAGE PRIORS (CONT.)

- Different choices of prior on ϕ_j and λ define different shrinkage priors:
 - Horseshoe: $\phi_j \sim C^+(0, 1), \lambda \sim C^+(0, 1)$
 - Normal-Gamma: $\phi_j^2 \sim \mathcal{G}(\vartheta, \lambda^2/2), \lambda^2 \sim \mathcal{G}(a_\lambda, b_\lambda)$
 - LASSO: special case of Normal-Gamma with $\vartheta = 1$
- **Efficient Gibbs sampler** available for all these variants [see Appendix](#)
- **Current gold standard** for continuous shrinkage in high-dimensional regression; priors adapt automatically to both sparse and dense signals
- Minimal input of the researcher required (tuning-free) and hence basically ML!
- Downside: No PIPs, but approximation possible with *“first shrink, then sparsify!”* (see, e.g., Huber, Koop, and Onorante, 2021 JBES)

SHRINKAGE PROFILES OF THE DIFFERENT PRIORS



A good shrinkage prior has a **spike at 0** (shrinks noise) and **heavy tails** (keeps large signals)

Application

When is growth at risk?

APPLICATION: WHEN IS GROWTH AT RISK?

PLAGBORG-MØLLER ET AL. (2020 *BROOKINGS PAPERS*)

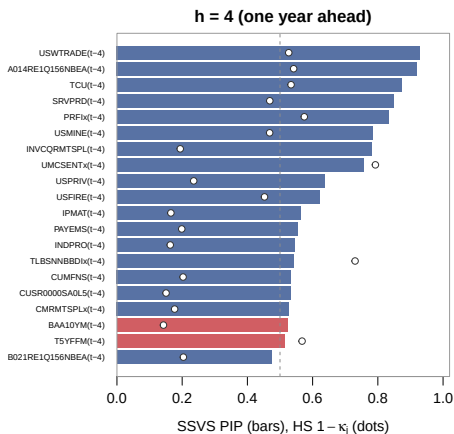
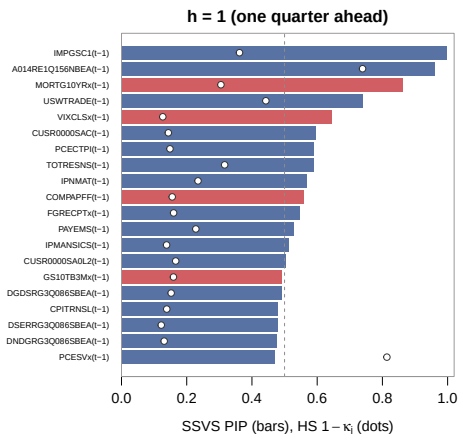
- Do financial variables predict (the **distribution** of) GDP growth?
- **Data**: FRED-QD augmented with key financial indicators (NFCI components); multi-country extension covering **13 advanced economies**
- Variable selection and shrinkage priors are **key**
- **Main finding**: Moments beyond the conditional mean are **poorly estimated**
 - Real indicators dominate (inventories, income, employment, housing, imports)
 - Only two financial variables enter robustly: implied volatility (VX0) and the AAA-to-Treasury spread (AAASPR)
- Financial variables carry **little predictive power** for tail risks

SETTING UP A SIMILAR VARIABLE SELECTION EXERCISE

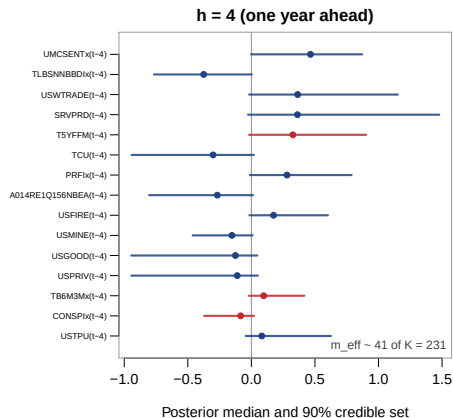
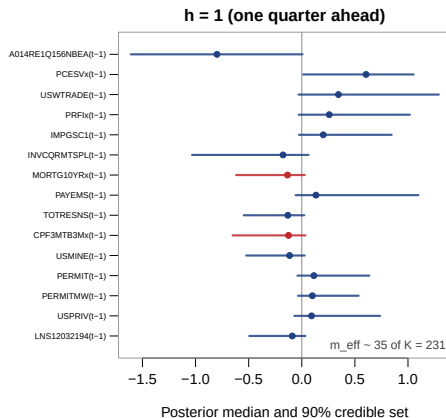
PLAGBORG-MØLLER ET AL. (2020, *BROOKINGS PAPERS*)

- Revisit the paper's US variable selection exercise with this session's tools
- Focus on FRED-QD and conditional mean
- **Data:** FRED-QD (2026-04 vintage); balanced panel of $K = 231$ standardised predictors (plus an intercept), 1973Q1–2019Q4 ($T = 187$ for $h = 1$) $\Rightarrow$ OLS infeasible
- **Regression setup**
 - Target: Annualised GDP growth, $y_t = \frac{400}{h} (\log \text{GDP}_t - \log \text{GDP}_{t-h})$ for $h \in \{1, 4\}$
 - K predictors $\mathbf{x}_{t-h}$ (including intercept)
- **Priors:** SSVS (gives PIPs) and horseshoe (gives shrinkage factors κ_j), with the Gibbs samplers sketched in the appendix (S02_BLR_Shrinkage.R file)

APPLICATION: WHICH PREDICTORS ARE SELECTED?



APPLICATION: FEW SIGNALS ESCAPE THE SHRINKAGE



Appendix

A noninformative prior

- Noninformative prior sets $\underline{a} = 0$ and $\underline{V}$ is big (implies large prior uncertainty)
- A common way: $\underline{V}^{-1} = \tau^{-1} \mathbf{I}_K$ where τ^{-1} is a scalar and let τ^{-1} go to zero
- This noninformative prior is improper and becomes:

$$p(\boldsymbol{\beta}, \sigma^2) \propto \frac{1}{\sigma^2}$$

- With this choice we get OLS results:

$$\boldsymbol{\beta} \mid \sigma^2, \mathbf{y} \sim \mathcal{N}(\bar{\boldsymbol{\beta}}, \sigma^2 \bar{\mathbf{V}})$$

$$\text{with } \bar{\mathbf{V}} = (\mathbf{X}'\mathbf{X})^{-1} \text{ and } \bar{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}$$

$$\sigma^2 \mid \mathbf{y} \sim \mathcal{IG}(\bar{a}, \bar{b})$$

$$\text{with } \bar{a} = \frac{T}{2} \text{ and } \bar{b} = \frac{1}{2} (\nu s^2)$$

Popular MCMC algorithms

The Gibbs sampler

- The **Gibbs sampler** uses conditional posteriors to draw $\beta^{(s)}$ and $(\sigma^2)^{(s)}$ for $s = 1, \dots, S$
- Averages yield posterior estimates, as with Monte Carlo integration
- Powerful and widely-used tool for posterior simulation in econometric models
- **General setup:** θ is a p -vector of parameters with likelihood $p(\mathbf{y} \mid \theta)$, prior $p(\theta)$ and posterior $p(\theta \mid \mathbf{y})$
- Partition θ into B **blocks**:

$$\theta = \left(\theta'_{(1)}, \theta'_{(2)}, \dots, \theta'_{(B)} \right)'$$

- For example, regression model: $B = 2$ with $\theta_{(1)} = \beta$ and $\theta_{(2)} = \sigma^2$

Intuition of the Gibbs sampler

- **Monte Carlo integration:** Draw from $p(\boldsymbol{\theta} \mid \mathbf{y})$ and average to estimate $\mathbb{E}[g(\boldsymbol{\theta}) \mid \mathbf{y}]$ for any function $g(\boldsymbol{\theta})$
- **Problem:** Drawing directly from $p(\boldsymbol{\theta} \mid \mathbf{y})$ is often **not possible** (e.g., not available in closed form)
- But it often is easy to draw from **full conditional posteriors**:

$$p(\boldsymbol{\theta}_{(b)} \mid \mathbf{y}, \{\boldsymbol{\theta}_{(j)}\}_{j \neq b}), \quad b = 1, \dots, B$$

- **Gibbs sampler:** Cycle through draws from each full conditional; in turn, produces a sequence $\boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)}, \dots, \boldsymbol{\theta}^{(S)}$ that can be averaged to estimate $\mathbb{E}[g(\boldsymbol{\theta}) \mid \mathbf{y}]$, just as with Monte Carlo integration

Gibbs sampler for Normal-Gamma

- ▶ The Normal-Gamma mixture representation enables **data augmentation** in a Gibbs sampler with well known conditionals
- ▶ For a regression with $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_T)$, iterate through:
 1. Draw $\boldsymbol{\beta} \mid \bullet \sim \mathcal{N}(\bar{\boldsymbol{\beta}}, \bar{\mathbf{V}})$, with $\bar{\mathbf{V}} = (\underline{\mathbf{V}}^{-1} + \sigma^{-2} \mathbf{X}'\mathbf{X})^{-1}$, $\bar{\boldsymbol{\beta}} = \sigma^{-2} \bar{\mathbf{V}} \mathbf{X}'\mathbf{y}$, and $\underline{\mathbf{V}} = \lambda^2 \text{diag}(\phi_1^2, \dots, \phi_K^2)$
 2. Draw $\sigma^2 \mid \bullet \sim \mathcal{IG}(\bar{a}, \bar{b})$, with $\bar{a} = \underline{a} + \frac{T}{2}$ and $\bar{b} = \underline{b} + \frac{1}{2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$
 3. Draw $\phi_j^2 \mid \bullet \sim \mathcal{GIG}(\vartheta - \frac{1}{2}, \lambda^2, \beta_j^2)$ for each $j = 1, \dots, K$
 4. Draw $\lambda^2 \mid \bullet \sim \mathcal{G}(a_\lambda + K\vartheta, b_\lambda + \frac{1}{2} \sum_{j=1}^K \phi_j^2)$

Gibbs sampler for the Horseshoe

- ▶ The **inverse-Gamma augmentation** of Makalic & Schmidt, 2016, *IEEE* (auxiliaries v_j, ξ) results in a Gibbs sampler with well known conditionals
- ▶ For $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, iterate through:
 1. Draw $\boldsymbol{\beta} \mid \bullet \sim \mathcal{N}(\bar{\boldsymbol{\beta}}, \bar{\mathbf{V}})$, with $\bar{\mathbf{V}} = (\underline{\mathbf{V}}^{-1} + \sigma^{-2}\mathbf{X}'\mathbf{X})^{-1}$, $\bar{\boldsymbol{\beta}} = \sigma^{-2}\bar{\mathbf{V}}\mathbf{X}'\mathbf{y}$, and $\underline{\mathbf{V}} = \lambda^2 \text{diag}(\phi_1^2, \dots, \phi_K^2)$
 2. Draw $\sigma^2 \mid \bullet \sim \mathcal{IG}(\bar{a}, \bar{b})$, with $\bar{a} = \underline{a} + \frac{T}{2}$ and $\bar{b} = \underline{b} + \frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$
 3. Draw $\phi_j^2 \mid \bullet \sim \mathcal{IG}\left(1, \frac{1}{v_j} + \frac{\beta_j^2}{2\lambda^2}\right)$ and $v_j \mid \phi_j^2 \sim \mathcal{IG}\left(1, 1 + \frac{1}{\phi_j^2}\right)$ for each j
 4. Draw $\lambda^2 \mid \bullet \sim \mathcal{IG}\left(\frac{K+1}{2}, \frac{1}{\xi} + \sum_{j=1}^K \frac{\beta_j^2}{2\phi_j^2}\right)$ and $\xi \mid \lambda^2 \sim \mathcal{IG}\left(1, 1 + \frac{1}{\lambda^2}\right)$